Parent Guide — primary / module-11-multiplication-near-base

IKS Curriculum

Table of Contents

Parent Guide — Module 11: Multiply-Near-Ten (Towards Nikhilam)

Your child's age: Primary (Grades 3–5) · **Module length:** 10 sessions × 30 min

What your child is learning

A finger trick for multiplying any two digits from 6 to 9. It's quick, mental, and once it clicks, your child will love showing it to you.

How it works:

> To multiply `7 × 8`: - On the LEFT hand, fold down (7 − 5) = 2 fingers. - On the RIGHT hand, fold down (8 − 5) = 3 fingers. - **Fingers DOWN on both hands** = 5. That's the TENS → **50**. - **Fingers UP on left × Fingers UP on right** = 3 × 2 = 6. That's the UNITS → **6**. - Total = **56**. ✓

It works for any pair from 6×6 to 9×9. Once your child has the trick, they'll know all these without memorising the times table.

The connection to older-grade math: This finger trick is exactly the same as the Middle-band "*Nikhilam* method" — the trick uses the fingers as a mnemonic for what older students call *deficits from 10*. Your child will meet the full version in Grade 6+.

What to expect this fortnight

A daily chant (the same one as Module 5's "Make 10"): > "1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!"

The finger trick (Day 3 onwards): Your child will multiply by holding their hands up and folding fingers.

A dot-pattern drawing (Day 4): Your child draws a grid of dots to *see* why 7 × 8 = 56.

A 1-minute Nikhilam demo (Day 6): The teacher will solve `97 × 96 = 9312` in under 5 seconds. Your child will say "wow." That's the point.

A creative project (Days 7–10): Your child designs a poster or 4-panel comic that teaches the finger trick to a younger child. Posters go on the wall on Day 10.

How to help — and how NOT to help

Helpful: - Let your child show you the finger trick. Ask them to multiply 7 × 8 in front of you. Their pride is the learning. - 5-minute home game: you say a multiplication (e.g. "8 × 7"), they answer using the finger trick. Try to beat their time. - For their poster: provide A4 paper and colour crayons. Let them design.

Not helpful: - Insisting they "just memorise the times table." Both ways work. The finger trick is a backup that *also* teaches them why multiplication works. - Doing their poster for them. Stick figures are fine; the math is the point. - Stressing about speed. Speed comes from practice.

A note on "Vedic mathematics"

You may have heard the term "Vedic mathematics." Here is the honest framing we use: - The finger trick is **folk math** — taught in many cultures and many schools. - The Sanskrit name *Nikhilam* comes from Bharati Krishna Tirthaji's *Vedic Mathematics* (1965, posthumous). - The math is real. The Sanskrit name is real. The claim that this specific technique comes from an ancient Vedic text is debated by historians.

At this age band we keep it simple: "It's a useful Indian trick with a Sanskrit name." When your child is older (Middle band), they'll learn the full algorithm; in Senior band, they'll read the scholarly debate.

What success looks like

By the end of the module, your child can: - Recall all 9 deficit pairs (1↔9, 2↔8, etc.) instantly. - Multiply any two digits from 6 to 9 using the finger trick. - Explain (with the dot-pattern drawing) why the trick works. - Show classmates the poster they designed.

Want to go deeper?

The Middle-band version (Grade 6–8) teaches the full Nikhilam method on 2-digit and 3-digit numbers near 100 and 1000. Preview: `curriculum/middle/module-11-multiplication-near-base/README.md`.

10-Day Lesson Plan

Module 11 (Primary) — 10-Day Lesson Plan

At a glance (30-min sessions)

Day	Title	Activity	Output
1	Make the Twin Game	Pair card game	First-day fun

Day	Title	Activity	Output
2	Twins Chant	Memorise the 9 pairs	Chant + recall
3	The Finger Trick	Hand multiplication 6×6 to 9×9	First mental answer
4	Why the Trick Works	Dot-pattern drawing of 7 × 8 = 56	Drawing pasted
5	Pairs Race	Speed game with finger trick	Personal best
6	Meet Nikhilam	Teacher demo of 97 × 96	Awe
7	Design a Game	Children design own poster / game	Sketch
8	Project Day 1	Build the poster	Poster halfway
9	Project Day 2	Finish + colour	Poster ready
10	Showcase	Wall display + 1-min share	Stickers

Daily structure (30 min)

Time	What
5 min	Settle + the "Twins" chant
7 min	Story / new game introduction
13 min	Game / activity
5 min	Sharing / reset

The daily chant

> "1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!"

(Same as Module 5's "Make 10" chant — reuse it.)

Sing it / chorus it with claps. By Day 3 it should be effortless.

Pacing notes

- Games > worksheets at this band. If a worksheet feels slow, drop it and add another game round.
- Day 6 (Nikhilam demo) is a "wow" moment — keep it short and mysterious.
- Days 8–10 build to the poster showcase. Confirm pair groupings by Day 6.

What children leave with

- Instant recall of all 9 deficit pairs.
- A finger trick they can use to multiply any two digits from 6 to 9.
- Curiosity about the older-grade trick (Nikhilam).
- One poster or comic they made and shared with classmates.

Homework

Homework — Multiplication Near a Power of 10 — Nikhilam (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — Finger-Trick Prelude** · Draw about today's concept. (~5 min)
2. **Day 2 — Two-Digit × Two-Digit Near 100 (Below)** · Draw about today's concept. (~5 min)
3. **Day 3 — Two-Digit × Two-Digit Near 100 (Above)** · Draw about today's concept. (~5 min)
4. **Day 4 — Mixed — One Above, One Below** · Draw about today's concept. (~5 min)
5. **Day 5 — 3-Digit × 3-Digit Near 1000** · Draw about today's concept. (~5 min)
6. **Day 6 — Why It Works — The Algebra** · Draw about today's concept. (~5 min)
7. **Day 7 — When To Use It** · Draw about today's concept. (~5 min)
8. **Day 8 — Speed Sprints** · Draw about today's concept. (~5 min)
9. **Day 9 — Mixed Practice + Module 7 Crossover** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.